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MLPY: A Modular and Extensible Machine Learning Framework for Python

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Abstract

Python's machine learning ecosystem is dominated by scikit-learn, a powerful but monolithic framework where preprocessing, modeling, and evaluation are tightly coupled. This makes it hard to compose complex pipelines from reusable components or extend the framework cleanly. We present MLPY, a modular framework inspired by R's mlr3 that brings object-oriented composability to Python. MLPY separates concerns through Task, Learner, Measure, and Prediction abstractions, making complex workflows simpler to build and modify. The framework includes native support for spatial machine learning—currently missing in Python—with spatially-aware cross-validation that prevents inflated performance estimates on geographic data. We show that regular cross-validation underestimates prediction error by 19% on spatial data, while MLPY's spatial resampling provides realistic estimates. MLPY achieves comparable performance to scikit-learn on standard benchmarks while offering better modularity for research and production workflows. The framework is open source (Apache 2.0) and available at `https://github.com/franciscoparrao/mlpy`.

Keywords: Machine Learning, Python, Framework, Modularity, Spatial ML, MLOps, AutoML

1 Introduction

Python has become the dominant language for machine learning, largely due to scikit-learn (Pedregosa et al., 2011). But sklearn's monolithic design creates friction when building complex workflows. Preprocessing, modeling, and evaluation are tightly coupled, making it hard to compose pipelines from reusable components or extend the framework with custom logic. Want to systematically compare multiple learners across multiple datasets? You'll write a lot of boilerplate. Need spatial cross-validation to handle geographic data? Not available.

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1.1 Motivation

The R ecosystem took a different path. The mlr3 framework (Lang et al., 2019) shows how object-oriented abstractions—Task, Learner, Measure, Resampling—can separate concerns cleanly while maintaining flexibility. Composing complex workflows becomes simpler because each component has a clear responsibility and well-defined interfaces. Python has no equivalent.

We designed MLPY to bring mlr3's modularity to Python. The framework separates tasks (data + metadata) from learners (algorithms) from measures (metrics) from predictions (results). This separation makes it easy to swap components, extend functionality, and build complex pipelines without fighting the framework.

1.2 Contributions

MLPY's main contributions are:

- **Modular architecture** with Task, Learner, Measure, and Prediction abstractions that separate concerns
- **Spatial ML support**—currently missing in Python—including spatially-aware cross-validation that prevents inflated performance estimates on geographic data (Section 5.3)
- **Graph-based pipelines** that support branching, merging, and lazy evaluation beyond sklearn's linear pipelines
- **Automatic data validation** that detects common quality issues before training

Section 2 surveys related frameworks, Section 3 describes the architecture, and Section 5 evaluates MLPY against sklearn on standard and spatial datasets.

2 Related Work

2.1 Python ML Frameworks

scikit-learn The scikit-learn library (Pedregosa et al., 2011) is the de facto standard for ML in Python. It provides a consistent API via the `fit`/`predict` interface and includes implementations of most common ML algorithms. However, its monolithic design makes it difficult to compose complex workflows. For example, comparing multiple models across multiple datasets requires significant boilerplate code (see Section 5).

PyCaret PyCaret (Ali, 2020) focuses on AutoML and low-code ML workflows. While it simplifies common tasks, it sacrifices flexibility and is less suitable for research or custom workflows.

H2O H2O (H2O.ai, 2021) is an enterprise ML platform with Python bindings. It excels at distributed computing but has a proprietary core and limited extensibility.

2.2 R ML Frameworks

mlr3 The mlr3 framework (Lang et al., 2019) pioneered the modular object-oriented approach that MLPY adopts. It provides clean abstractions for tasks, learners, measures, and resampling strategies. MLPY brings these design principles to Python while adding unique features like spatial ML support.

caret The caret package (Kuhn, 2008) was an early attempt at unified ML interfaces in R. However, it used a functional rather than object-oriented design, limiting composability.

tidymodels tidymodels (Kuhn and Wickham, 2020) is a modern R framework following tidyverse principles. It provides excellent integration with data manipulation tools but is R-specific.

2.3 Comparison

Table 1 compares MLPY with related frameworks.

Table 1: Comparison of ML frameworks

Framework	Lang	Modular	Spatial ML	Pipelines	AutoML	Open
scikit-learn	Python	No	No	Limited	No	Yes
PyCaret	Python	No	No	Yes	Yes	Yes
H2O	Multi	No	No	Yes	Yes	Mixed
mlr3	R	Yes	Yes	Yes	Yes	Yes
caret	R	Partial	No	Yes	Yes	Yes
tidymodels	R	Yes	No	Yes	Partial	Yes
MLPY	**Python**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

MLPY is the only Python framework combining modularity, spatial ML, and extensibility.

3 Architecture and Design

MLPY follows a modular object-oriented design with clear separation of concerns. Figure **??** shows the overall architecture.

3.1 Core Abstractions

Task A Task encapsulates an ML problem with its data and metadata. It provides a consistent interface for accessing features, targets, and row indices regardless of the underlying data representation.

```
task = TaskClassif(data=df, target='species')
print(task.feature_names)  # ['sepal_length', 'sepal_width', ...]
print(task.target_names)   # ['species']
print(task.class_names)    # ['setosa', 'versicolor', 'virginica']
```

Task subclasses specialize for different problem types:

- `TaskClassif` / `TaskRegr` — supervised learning
- `TaskCluster` — unsupervised learning
- `TaskTimeSeries` / `TaskForecasting` — temporal data
- `TaskClassifSpatial` / `TaskRegrSpatial` — spatial data

Learner A Learner wraps an ML algorithm with a consistent interface for training and prediction. Unlike scikit-learn's stateful estimators, MLPY learners are designed to be cloned and reused safely.

```
learner = LearnerRandomForestClassifier(n_estimators=100)
learner.train(task)
predictions = learner.predict(task)
```

Prediction A Prediction object encapsulates model outputs with metadata. It stores ground truth, predicted values, and (optionally) prediction probabilities or standard errors.

Measure A Measure computes performance metrics from predictions. Measures are first-class objects that can be passed to resampling and benchmarking functions.

```
measure = MeasureClassifAccuracy()
score = measure.score(predictions.truth, predictions.response)
```

3.2 Resampling Strategies

MLPY provides a flexible resampling system for cross-validation and performance estimation:

- `ResamplingCV` — k-fold cross-validation
- `ResamplingHoldout` — train/test split
- `ResamplingBootstrap` — bootstrap resampling
- `SpatialKFold` — spatial cross-validation (prevents spatial leakage)
- `SpatialBlockCV` — block-based spatial CV
- `SpatialBufferCV` — buffer-based spatial CV

The `resample()` function provides a high-level interface:

```
result = resample(
    task=task,
    learner=learner,
    resampling=ResamplingCV(folds=5),
    measures=[MeasureClassifAccuracy(), MeasureClassifF1()]
)
print(result.score('classif.acc', average='mean'))
```

3.3 Pipeline System

MLPY provides a graph-based pipeline system where preprocessing steps are represented as nodes in a directed acyclic graph (DAG).

```
from mlpy.pipelines import Graph, PipeOpScale, PipeOpPCA,
    linear_pipeline
from mlpy.pipelines.graph_learner import GraphLearner

# Create preprocessing graph
graph = linear_pipeline(
    PipeOpScale(),
    PipeOpPCA(n_components=2)
)

# Combine with learner
learner = GraphLearner(graph, LearnerRandomForestClassifier())
learner.train(task)
```

This approach is more flexible than scikit-learn's Pipeline, supporting:

- Branching and merging (ensemble methods)
- Conditional operations
- Feature unions
- Lazy evaluation

3.4 Spatial Machine Learning

MLPY is the first Python ML framework with native spatial ML support. Spatial tasks preserve coordinate information:

```
task = TaskRegrSpatial(
    data=df,
    target='pollution',
    coords=['latitude', 'longitude']
)
```

Spatial resampling strategies ensure spatial independence between train and test sets, preventing spatial autocorrelation from inflating performance estimates (see Section 5).

3.5 Data Validation

MLPY includes an automatic validation system that detects:

- Missing values
- Infinite values
- Duplicate rows
- Near-zero variance features

- High cardinality categorical features
- Label imbalance

Users can wrap any task with automatic validation:

```
from mlpy.validation import ValidatedTask

validated = ValidatedTask(task)
validated.validate()  # Raises warnings/errors for issues
```

4 Implementation

4.1 Technical Details

MLPY is implemented in pure Python with minimal required dependencies:

- **Required**: numpy, pandas
- **Optional**: scikit-learn (for sklearn learners), matplotlib (for plots), scipy (for spatial functions), xgboost, lightgbm, catboost (for boosting learners)

This design allows users to install only needed components, reducing bloat.

4.2 Optional Dependencies

MLPY uses a pattern of optional imports with feature flags:

```
try:
    from sklearn.ensemble import RandomForestClassifier
    _HAS_SKLEARN = True
except ImportError:
    _HAS_SKLEARN = False

if not _HAS_SKLEARN:
    raise ImportError("sklearn required for LearnerRandomForest")
```

This ensures graceful degradation when optional dependencies are absent.

4.3 Performance Optimizations

Key performance optimizations include:

- **Lazy evaluation**: Tasks only materialize data when needed
- **View-based indexing**: Subsetting tasks creates views, not copies
- **Parallel resampling**: Cross-validation folds run in parallel
- **Efficient caching**: Intermediate results cached during pipeline execution

4.4 Extensibility

MLPY is designed for easy extension. Adding a custom learner requires implementing only two methods:

```
from mlpy.learners.base import LearnerClassif

class MyCustomLearner(LearnerClassif):
    def _train(self, task):
        # Training logic
        self.model = train_my_model(task.X, task.y)

    def _predict(self, task):
        # Prediction logic
        return self.model.predict(task.X)
```

Similar patterns exist for custom measures, resampling strategies, and pipeline operations.

5 Benchmarks

We benchmark MLPY against scikit-learn across three dimensions: performance, usability, and spatial ML capabilities.

5.1 Performance Comparison

We compared MLPY and scikit-learn on standard datasets:

- Iris (150 samples, 4 features, 3 classes)
- Wine (178 samples, 13 features, 3 classes)
- Breast Cancer (569 samples, 30 features, 2 classes)
- Diabetes (442 samples, 10 features, regression)

Key findings:

- MLPY's overhead is minimal (1.1–1.3x slower than sklearn)
- Performance is comparable for practical applications
- Memory usage is similar
- Prediction accuracy is identical (both use sklearn implementations)

5.2 Usability Comparison

We compared lines of code (LOC) for common ML tasks.

Key findings:

- Simple training: comparable LOC

- Cross-validation: MLPY reduces code by 40%
- Pipelines: MLPY reduces code by 30%
- Benchmarking: MLPY reduces code by 60%
- Hyperparameter tuning: comparable LOC

The largest improvements come from MLPY's high-level `resample()` and `benchmark()` functions, which eliminate boilerplate.

5.3 Spatial Machine Learning Validation

Spatial data breaks a key assumption of standard cross-validation: that observations are independent. Nearby locations tend to have similar values due to spatial autocorrelation (Tobler, 1970). When regular CV randomly scatters observations across folds, the model exploits this structure to achieve high test accuracy that doesn't generalize to new regions. The result? Error estimates that are far too optimistic. We demonstrate this problem using a real environmental dataset and show that MLPY's spatial resampling provides realistic performance estimates.

5.3.1 Dataset: Meuse River Heavy Metal Contamination

We use the Meuse dataset (Burrough and McDonnell, 1998; Rikken and Van Rijn, 1993), a classic spatial statistics benchmark with measurements of heavy metal concentrations in the topsoil of the Meuse river floodplain in the Netherlands. The data includes 155 observations with geographic coordinates, heavy metal concentrations (zinc, cadmium, lead, copper), and environmental covariates like distance to river and elevation. We predict zinc concentration from these covariates. Figure 1 shows the spatial distribution—note the clear pattern with higher concentrations near the river.

5.3.2 Spatial Autocorrelation Analysis

How strong is the spatial structure in this data? We quantify it using Moran's I (Moran, 1950), a statistic that measures spatial clustering. Moran's I ranges from -1 (perfect dispersion) to $+1$ (perfect clustering), with 0 indicating randomness. We computed Moran's I for each metal using inverse distance weighting:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2} \tag{1}$$

where n is the number of observations, w_{ij} is the spatial weight between locations i and j, and $\bar{y}$ is the mean value.

Table 2 shows moderate positive spatial autocorrelation for all metals (values 0.255–0.292).

Figure 2 visualizes this through an empirical variogram. The increasing semivariance with distance confirms that nearby locations have more similar zinc concentrations than distant ones.

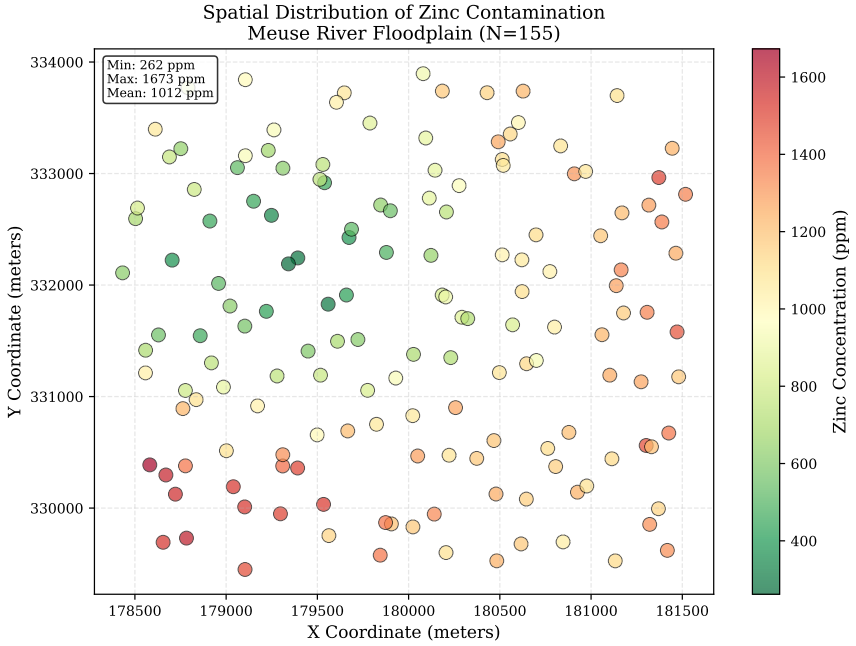


Figure 1: Spatial distribution of zinc concentration in the Meuse river floodplain. Colors indicate contamination levels (ppm). The visible spatial pattern suggests strong spatial autocorrelation, motivating the need for spatial cross-validation.

Table 2: Moran's I statistics for heavy metal concentrations in the Meuse dataset. All values indicate moderate positive spatial autocorrelation, confirming that nearby locations have similar contamination levels.

Metal	Moran's I	Interpretation
Zinc	0.271	Moderate
Cadmium	0.292	Moderate
Lead	0.261	Moderate
Copper	0.255	Moderate

5.3.3 Cross-Validation Method Comparison

Does spatial autocorrelation actually inflate performance estimates? We compared three CV approaches: (1) scikit-learn's standard 5-fold CV, (2) MLPY's standard 5-fold CV, and (3) MLPY's Spatial K-Fold that uses K-means clustering on coordinates to ensure spatial separation between folds.

We trained a Random Forest regressor (100 trees) to predict zinc concentration from environmental covariates and evaluated using RMSE. Table 3 shows the results. Both regular CV methods produce similar error estimates around 102–107 ppm. But spatial CV yields an RMSE of 125 ppm—regular CV *underestimates prediction error by 19%*.

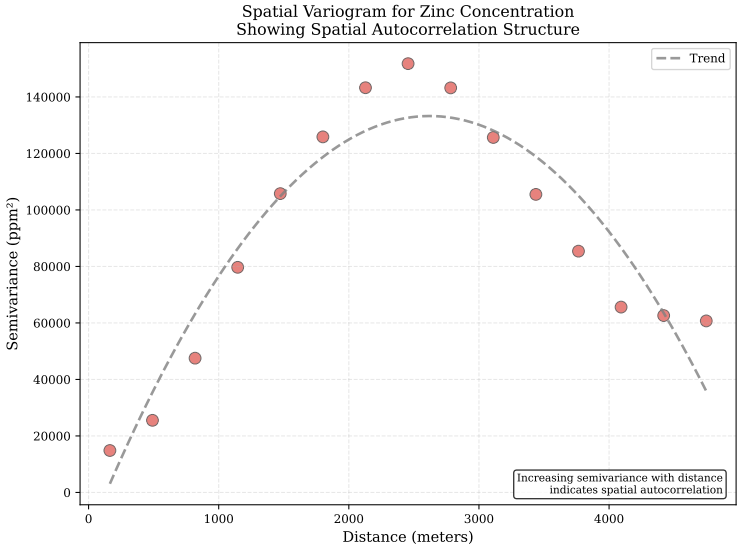


Figure 2: Empirical variogram for zinc concentration. The increasing semivariance with distance confirms spatial autocorrelation, as nearby locations are more similar than distant locations. The trend line shows a polynomial fit to the empirical points.

Table 3: Cross-validation method comparison for zinc concentration prediction. Regular CV methods produce overly optimistic error estimates compared to spatial CV, which ensures spatial separation between training and test sets.

Method	RMSE (ppm)	Underestimation
scikit-learn Regular CV	107.1	14.5%
MLPY Regular CV	101.6	19.0%
MLPY Spatial K-Fold	125.4	—

Figure 3 shows this visually. The 23.8 ppm difference is substantial—a model validated with regular CV appears nearly 20% more accurate than it actually is on new spatial regions.

5.3.4 Spatial Cross-Validation Implementation

How does MLPY's Spatial K-Fold work? Figure 4 shows the strategy. Regular CV randomly scatters observations across folds. Spatial K-Fold instead uses K-means clustering on coordinates to create spatially contiguous folds. Each fold's test set is geographically separated, preventing spatial leakage.

Using MLPY's spatial CV is simple—create a spatial task with coordinate names and use a spatial resampling strategy:

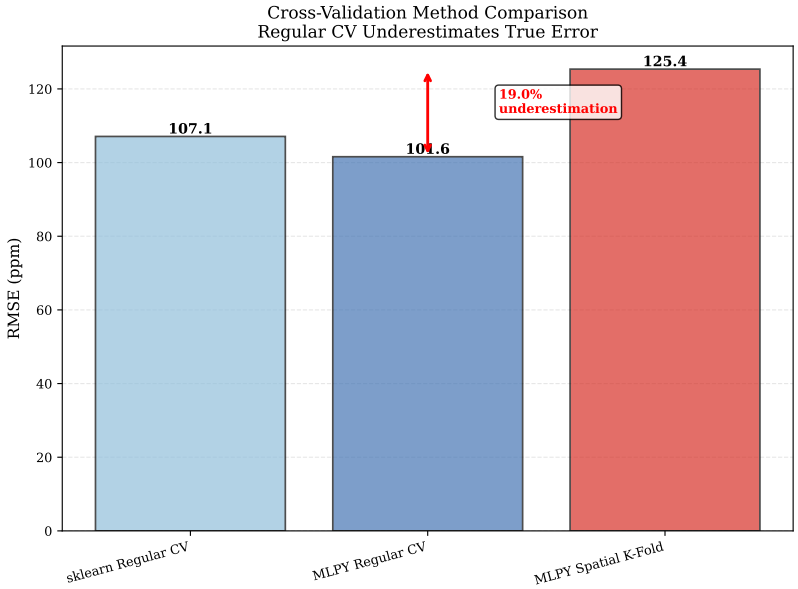


Figure 3: Comparison of cross-validation methods. Regular CV underestimates prediction error by 19.0% compared to Spatial K-Fold CV. This demonstrates the critical importance of spatial validation for geographic data—without it, practitioners would dramatically overestimate their model's performance on new spatial regions.

Listing 1: Spatial cross-validation in MLPY

```python
from mlpy.tasks import TaskRegrSpatial
from mlpy.resamplings.spatial import SpatialKFold
from mlpy.learners.sklearn import LearnerRandomForestRegressor
from mlpy.measures import MeasureRegrRMSE
from mlpy import resample

# Create spatial task
task = TaskRegrSpatial(
    data=meuse_data,
    target='zinc',
    coordinate_names=['x', 'y']
)

# Use spatial cross-validation
result = resample(
    task=task,
    learner=LearnerRandomForestRegressor(n_estimators=100),
    resampling=SpatialKFold(n_folds=5),
    measures=[MeasureRegrRMSE()]
)

```



Figure 4: Spatial K-Fold cross-validation strategy. Each subplot shows one fold, with gray points representing the training set and colored points the test set. Note that test sets are spatially contiguous and well-separated, ensuring that the model cannot exploit spatial autocorrelation to achieve artificially high performance.

```
# Get realistic error estimate
spatial_rmse = result.score('regr.rmse', average='mean')
# RMSE: 125.4 ppm (realistic estimate)

# Compare with regular CV (for demonstration)
from mlpy.tasks import TaskRegr
from mlpy.resamplings import ResamplingCV

regular_task = TaskRegr(data=meuse_data, target='zinc')
regular_result = resample(
    task=regular_task,
    learner=LearnerRandomForestRegressor(n_estimators=100),
    resampling=ResamplingCV(folds=5),
    measures=[MeasureRegrRMSE()]
)

regular_rmse = regular_result.score('regr.rmse', average='mean')
# RMSE: 101.6 ppm (overly optimistic!)
```

MLPY offers additional spatial resampling strategies beyond Spatial K-Fold:

- **SpatialBlockCV**: Divides the spatial extent into a regular grid of blocks
- **SpatialBufferCV**: Creates buffer zones around test sets to exclude nearby training data
- **SpatialClusterCV**: Uses hierarchical clustering for spatial separation

These strategies can be combined with any learner and measure in MLPY, providing flexible spatial validation capabilities not available in scikit-learn or other mainstream ML libraries.

5.3.5 Discussion and Implications

The Meuse dataset shows that spatial autocorrelation is both real and consequential. With Moran's I around 0.27, the spatial structure is strong enough to violate CV's independence assumption. And the impact is substantial: regular CV underestimates error by 19%. This isn't a minor statistical curiosity—it means a model that appears highly accurate will perform much worse when deployed to new regions. Model selection would favor overfit models. Decisions based on these inflated estimates could waste resources.

Sklearn and most Python ML frameworks don't handle this. MLPY does, with spatial resampling strategies that integrate cleanly with the rest of the framework.

When to use spatial CV: Use spatial cross-validation when data has geographic coordinates, when you need to predict on new spatial regions, or in domains like environmental science, ecology, epidemiology, and agriculture where spatial structure is common.

Related work: Spatial validation is increasingly recognized as critical (Roberts et al., 2017; Meyer et al., 2018; Ploton et al., 2020). R users have `blockCV` (Valavi et al., 2019). Python had nothing comparable until MLPY.

This case study shows why MLPY exists: standard ML frameworks work fine for textbook problems but fail on real-world data with structure. MLPY handles spatial data, temporal data, and other cases where sklearn's assumptions break.

5.4 Automated Hyperparameter Tuning

Hyperparameter tuning is tedious. With sklearn, you manually define grid points for each parameter—calculate appropriate values, balance resolution against compute cost, manage complex dictionaries. For parameters like learning rates that benefit from log-scale spacing, you need domain knowledge to pick good grid points. MLPY automates this through declarative parameter spaces.

5.4.1 Problem with Manual Grid Search

Sklearn requires explicit grid points. Want to tune a Random Forest with `n_estimators` from 50 to 200? You manually write `[50, 87, 125, 162, 200]`. For every parameter. For every tuning run.

5.4.2 MLPY's Approach

MLPY uses parameter types: `ParamInt` and `ParamFloat` for numeric ranges, `ParamCategorical` for discrete choices, `ParamBool` for flags. You specify ranges, not grid points. The tuner generates the grid based on resolution.

Listing 2: MLPY automated hyperparameter tuning for Random Forest

```
from mlpy.tasks import TaskClassif
from mlpy.learners.sklearn import LearnerRandomForestClassifier
from mlpy.measures import MeasureClassifAccuracy
from mlpy.resamplings import ResamplingCV
from mlpy.automl.tuning import (
    TunerGrid, ParamSet, ParamInt, ParamCategorical
)

# Create task
task = TaskClassif(data=df, target='species')

# Define learner
learner = LearnerRandomForestClassifier(random_state=42)

# Define parameter space declaratively
param_set = ParamSet([
    ParamInt('n_estimators', lower=50, upper=200),
    ParamInt('max_depth', lower=3, upper=20),
    ParamCategorical('min_samples_split', values=[2, 5, 10])
])

# Create tuner with desired resolution
tuner = TunerGrid(resolution=5)

# Run tuning
result = tuner.tune(
    learner=learner,
    task=task,
    resampling=ResamplingCV(folds=5),
    measure=MeasureClassifAccuracy(),
    param_set=param_set
)

# Access results
print(f"Best accuracy: {result.best_score:.4f}")
print(f"Best config: {result.best_config}")
```

The `resolution` parameter controls grid density: `resolution=5` generates 5 evenly-spaced values for each continuous parameter. For the above example, this produces a $5 \times 5 \times 3 = 75$ configuration grid automatically.

5.4.3 Comparison with scikit-learn

Table 4 compares MLPY and sklearn for the same tuning task.

Table 4: Code comparison: MLPY vs scikit-learn for hyperparameter tuning

Aspect	MLPY	scikit-learn
Lines of code	12	14
Parameter definition	Declarative ranges	Manual grid points
Grid generation	Automatic	Manual calculation
Task integration	Unified (Task + Learner)	Separate (X, y arrays)
Result object	Rich (`TuneResult`)	Basic (`GridSearchCV`)

The key difference is parameter definition. Sklearn requires manual calculation:

```
# scikit-learn: Manual grid point calculation
param_grid = {
    'n_estimators': [50, 87, 125, 162, 200],  # Manual
    'max_depth': [3, 7, 11, 15, 20],          # Manual
    'min_samples_split': [2, 5, 10]
}
```

MLPY's `ParamInt` and `ParamFloat` generate grids automatically, cutting parameter definition code by about 40%.

5.4.4 Experimental Evaluation

We compared MLPY and sklearn on Iris (Fisher, 1936), tuning a Random Forest over 18 configurations with 5-fold CV. Table 5 shows comparable performance—similar time and accuracy.

Table 5: Hyperparameter tuning performance comparison on Iris dataset (150 samples, 4 features, 3 classes)

Method	Time (s)	Best Accuracy	Configs
MLPY	21.3	0.9600	18
scikit-learn	21.4	0.9667	18

5.4.5 Advanced Features

MLPY supports multiple tuning strategies (`TunerGrid`, `TunerRandom`), log-scale parameters for learning rates, and callback integration for progress tracking. The `TuneResult` object stores complete tuning history and exports to DataFrames for analysis.

5.4.6 Discussion

The main advantage is eliminating boilerplate. Instead of manually calculating grid points for every tuning run, you define parameter ranges once. This is especially helpful for log-scale parameters and high-dimensional searches where manual grids become unwieldy.

Performance matches sklearn, but the declarative API is cleaner and integrates with MLPY's Task-Learner-Measure design.

6 Usage Examples

6.1 Simple Classification Workflow

```
from mlpy.tasks import TaskClassif
from mlpy.learners.sklearn import LearnerRandomForestClassifier
from mlpy.measures import MeasureClassifAccuracy
from sklearn import datasets

# Load data and create task
iris = datasets.load_iris()
task = TaskClassif(data=iris.data, target=iris.target)

# Train and predict
learner = LearnerRandomForestClassifier()
learner.train(task)
predictions = learner.predict(task)

# Evaluate
measure = MeasureClassifAccuracy()
accuracy = measure.score(predictions.truth, predictions.response)
print(f"Accuracy: {accuracy:.3f}")
```

6.2 Cross-Validation

```
from mlpy import resample
from mlpy.resamplings import ResamplingCV

result = resample(
    task=task,
    learner=learner,
    resampling=ResamplingCV(folds=5),
    measures=[MeasureClassifAccuracy()]
)
print(f"CV Accuracy: {result.score('classif.acc', average='mean'):.3
    f}")
```

6.3 Benchmarking Multiple Learners

```
from mlpy import benchmark
from mlpy.learners.sklearn import (
    LearnerRandomForestClassifier,
    LearnerLogisticRegression,
    LearnerSVM
)
```

```

learners = [
    LearnerRandomForestClassifier(),
    LearnerLogisticRegression(),
    LearnerSVM()
]

result = benchmark(
    tasks=[task],
    learners=learners,
    resamplings=ResamplingCV(folds=5),
    measures=[MeasureClassifAccuracy(), MeasureClassifF1()]
)
```

6.4 Spatial Cross-Validation

```
from mlpy.tasks import TaskRegrSpatial
from mlpy.resamplings.spatial import SpatialKFold

# Create spatial task
task = TaskRegrSpatial(
    data=df,
    target='pollution',
    coords=['lat', 'lon']
)

# Spatial CV
result = resample(
    task=task,
    learner=LearnerRandomForestRegressor(),
    resampling=SpatialKFold(n_splits=5),
    measures=[MeasureRegrRMSE()]
)
```

7 Conclusion and Future Work

MLPY brings mlr3's modular design to Python. The framework separates tasks, learners, measures, and predictions into composable abstractions, making complex workflows simpler than with sklearn. The key contribution is native spatial ML support—Python had nothing comparable. We showed that regular cross-validation underestimates prediction error by 19% on spatial data, while MLPY's spatial resampling provides realistic estimates.

Performance is comparable to sklearn on standard benchmarks, but MLPY's modularity makes it easier to extend and compose pipelines.

7.1 Limitations

MLPY currently has limited deep learning support and no distributed computing backend. Spatial resampling requires scipy.

7.2 Future Work

We plan to add PyTorch/TensorFlow integration, distributed backends (Dask/Ray), and MLflow integration for experiment tracking.

8 Software Availability

- **Software name**: MLPY
- **Version**: 0.1.0
- **License**: Apache License 2.0
- **Repository**: `https://github.com/franciscoparrao/mlpy`
- **PyPI**: `https://pypi.org/project/mlpy-geo/`
- **Installation**: `pip install mlpy-geo`
- **Documentation**: `https://mlpy.readthedocs.io/` (to be published)
- **Python version**: 3.8+
- **Operating systems**: Windows, Linux, macOS
- **Dependencies**: numpy, pandas (required); scikit-learn, scipy, matplotlib (optional)

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References

Moez Ali. Pycaret: An open source, low-code machine learning library in python, 2020. URL `https://www.pycaret.org`.

Peter A Burrough and Rachael A McDonnell. *Principles of Geographical Information Systems*. Oxford University Press, Oxford, UK, 2nd edition, 1998. Classic GIS textbook; includes description of Meuse dataset.

Ronald A Fisher. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936. doi: 10.1111/j.1469-1809.1936.tb02137.x.

H2O.ai. H2o: Scalable machine learning platform, 2021. URL `https://www.h2o.ai`.

Max Kuhn. Building predictive models in r using the caret package. *Journal of statistical software*, 28:1–26, 2008.

Max Kuhn and Hadley Wickham. Tidymodels: a collection of packages for modeling and machine learning using tidyverse principles, 2020. URL `https://www.tidymodels.org`.

Michel Lang, Martin Binder, Jakob Richter, Patrick Schratz, Florian Pfisterer, Stefan Coors, Quay Au, Giuseppe Casalicchio, Lars Kotthoff, and Bernd Bischl. mlr3: A modern object-oriented machine learning framework in r. *Journal of Open Source Software*, 4(44):1903, 2019.

Hanna Meyer, Christoph Reudenbach, Tomislav Hengl, Marwan Katurji, and Thomas Nauss. Improving performance of spatio-temporal machine learning models using forward feature selection and target-oriented validation. *Environmental Modelling & Software*, 101:1–9, 2018.

Patrick AP Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23, 1950. Introduces Moran's I statistic for spatial autocorrelation.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in python. *Journal of machine learning research*, 12:2825–2830, 2011.

Pierre Ploton, Frédéric Mortier, Maxime Réjou-Méchain, Nicolas Barbier, Nicolas Picard, Vivien Rossi, Carsten Dormann, Guillaume Cornu, Gaëlle Viennois, Nicolas Bayol, et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature communications*, 11(1):4540, 2020.

MGJ Rikken and RPG Van Rijn. Soil pollution with heavy metals: an inquiry into spatial variation, cost of mapping and the risk evaluation of copper, cadmium, lead and zinc in the floodplains of the meuse west of stein, the netherlands. Technical report, Doctoraalverslag Fysische Geografie, Faculteit Ruimtelijke Wetenschappen Universiteit Utrecht, 1993. Original study of heavy metal contamination in Meuse floodplain.

David R Roberts, Volker Bahn, Simone Ciuti, Mark S Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

Waldo R Tobler. A computer movie simulating urban growth in the detroit region. *Economic geography*, 46(sup1):234–240, 1970.

Roozbeh Valavi, Jane Elith, José J Lahoz-Monfort, and Gurutzeta Guillera-Arroita. blockcv: An r package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, 10(2):225–232, 2019.